

Common RuntimeExceptions in Java

NullPointerException (NPE)

Occurs when you try to use null where an object is required.

```
String s = null;  
int len = s.length(); // NullPointerException
```

ArrayIndexOutOfBoundsException

Occurs when accessing an array with an invalid index.

```
int[] arr = {1, 2, 3};  
int n = arr[5]; // ArrayIndexOutOfBoundsException
```

StringIndexOutOfBoundsException

Occurs when accessing a character in a string with an invalid index.

```
String s = "Java";  
char c = s.charAt(10); // StringIndexOutOfBoundsException
```

ArithmeticException

Occurs when performing illegal arithmetic operations, e.g., division by zero.

```
int x = 10 / 0; // ArithmeticException
```

ClassCastException

Occurs when an object is cast to a subclass it is not an instance of.

```
Object obj = "Hello";  
Integer num = (Integer) obj; // ClassCastException
```

NumberFormatException

Occurs when parsing a string that is not a valid number.

```
String s = "abc";  
int n = Integer.parseInt(s); // NumberFormatException
```

IllegalArgumentException

Occurs when a method receives an invalid argument.

```
Thread t = new Thread();  
t.setPriority(20); // IllegalArgumentException
```

IllegalStateException

Occurs when a method is invoked at the wrong time or in the wrong state.

```
Iterator<String> it = List.of("A", "B").iterator();  
it.remove(); // IllegalStateException
```

UnsupportedOperationException

Occurs when an operation is not supported, e.g., modifying an immutable collection.

```
List<String> list = List.of("A", "B");  
list.add("C"); // UnsupportedOperationException
```

ConcurrentModificationException

Occurs when a collection is modified while iterating with an iterator.

```
List<String> list = new ArrayList<>(List.of("A", "B"));  
for (String s : list) {  
    list.add("C"); // ConcurrentModificationException  
}
```

IndexOutOfBoundsException

Superclass for exceptions when index is out of range in lists/arrays.

```
List<Integer> nums = new ArrayList<>();  
nums.get(5); // IndexOutOfBoundsException
```

NegativeArraySizeException

Occurs when creating an array with a negative size.

```
int[] arr = new int[-5]; // NegativeArraySizeException
```

SecurityException

Occurs when a denied security access is attempted.

```
System.setSecurityManager(new SecurityManager());  
System.exit(0); // SecurityException
```

MissingResourceException

Occurs when a required resource (like a properties file) is missing.

```
ResourceBundle.getBundle("nonexistent"); // MissingResourceException
```

ArrayStoreException

Occurs when storing the wrong type in an array.

```
Object[] arr = new String[2];  
arr[0] = 10; // ArrayStoreException
```